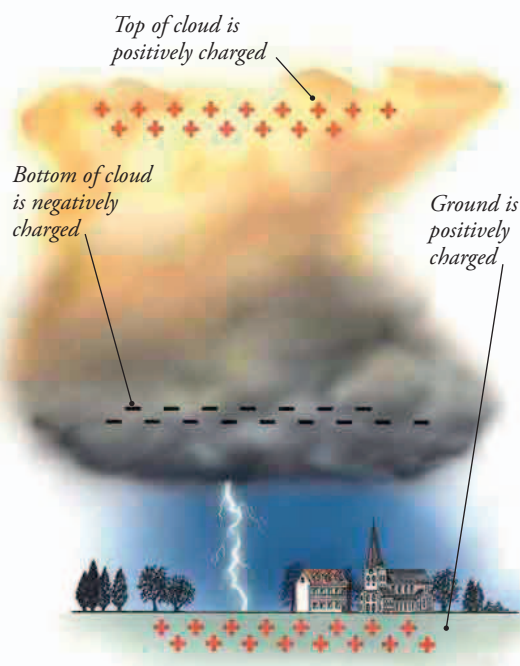




SUPERCHARGED

When ice particles are hurled up and down inside a cloud, they rub against each other. This generates electricity, charging up the cloud like a giant battery. The bottom of the cloud becomes negatively charged, while the top and the ground underneath become positively charged. If the charges grow big enough, lightning arcs from the cloud to the ground.



▲ ELECTRIFIED CLOUD

The negative charge at the base of a thundercloud creates a positive charge in the ground below. This difference in charge causes lightning.

LIGHTNING STRIKES

A lightning strike starts when a highly charged, branched “leader” threads its way through the air to the ground. When it touches the ground, a huge charge shoots back up the leader in a bright flash.

Within a few thousandths of a second, the air in the path of the lightning is heated to 54,000°F (30,000°C). This makes the air expand explosively, causing the shock wave that we hear as thunder.

► FLASHBACK

One of the branched leaders of this lightning strike has touched the ground, triggering a dazzling return flash.

BOLT FROM THE BLUE

The intense heat generated by lightning often starts fires, which can spread to destroy forests and buildings. Tree trunks may have long scars burned in them from top to bottom by lightning. When desert sand is struck by lightning, the heat can melt the sand, fusing it together into a natural form of glass called a fulgurite.



► SOLID LIGHTNING

Lightning striking sand often forms tube-shaped fulgurites, like this one.